

BAYESIAN ESTIMATION IN CHANGE-POINT HAZARD RATE MODELS

P Groenewald & A. Schoeman

Department of Mathematical Statistics, University of the Orange Free State,
P.O. Box 339, Bloemfontein 9300, South Africa

Key words: Bathtub hazard rate; fractional Bayes factor; survival function.

Summary: In this paper the Bayesian estimation of the parameters in a hazard rate model with change-points is considered. Among the models considered are the exponential model with a single abrupt change in the hazard rate, models with non-increasing or non-decreasing hazard rates and the non-monotonic ‘bathtub’ hazard rate model. The change-point models are constructed by combining exponential and Weibull lifetime distributions in various ways. Model comparisons are performed by means of the fractional Bayes factor and an MCMC procedure of obtaining the fractional Bayes factor is described. The method is illustrated in two applications to lifetime data.

1. Introduction

An important concept in survival analysis and reliability theory is the hazard rate which can be interpreted as the instantaneous failure rate, or the conditional density of failure at time t , given survival until time t . It is defined as

$$\begin{aligned}h(t) &= f(t|T \geq t) \\ &= \frac{f(t)}{1 - F(t)}.\end{aligned}\tag{1.1}$$

where $F(t) = P[T \leq t]$ is the CDF of the density $f(t)$.

STMA (2001) SUBJECT CLASSIFICATION: 14:130(04:070)

Researchers are often interested in detecting changes in the hazard rate of some lifetime distribution. The most commonly used lifetime distribution, the exponential distribution, has constant hazard rate. The assumption of constant hazard rate seems unrealistic in many applications due to high infant mortality rate, ageing, or both, but may often be useful over a limited time span.

A number of papers deal with the problem of detecting the change-point in the hazard rate, assuming a constant hazard rate up to time τ and then another different constant thereafter. Classical approaches to testing and estimation for this situation include Matthews and Farewell (1982), Matthews, Farewell and Pyke (1985), Nguyen, Rogers and Walker (1984), Yao (1986) and Pham and Nguyen (1990, 1993).

A more general model is dealt with by Basu, Ghosh and Joshi (1988) and Ghosh and Joshi (1992). They assume a non-increasing continuous function for the hazard rate before the change-point and a constant rate thereafter. In Mitra and Basu (1995) a general methodology is proposed for consistent estimation of the change-point in models with non-monotonic hazard functions.

Kulasekera and Saxena (1991) consider 'bathtub' and 'upside-down bathtub' shaped hazard functions and based their estimator of the change-point on kernel estimators of the density function.

The Bayesian approach to the problem of a single abrupt change in a constant hazard rate is used in Achcar and Bolfarine (1989) and Achcar and Loibel (1998). Ghosh *et al.* (1993, 1996) examine the asymptotics of a Bayesian approach for the same model, but under the assumption of a lower hazard rate after the change-point. Approximations to the posterior distribution of the change-point are also derived in Ghosal *et al.* (1999) under the slightly more general assumption of a non-increasing hazard rate before and after the

change-point. Zacks (1972) considers a constant failure rate (exponential) that changes at an unknown time point to an increasing (Weibull) rate.

In section 2 we treat the constant hazard rate with special reference to the Bayes factor in favour of a change-point model. In sections 4 and 5 Bayesian estimation under the assumptions of monotonic and non-monotonic hazard rates respectively are considered.

2. Detecting a change-point in a constant hazard rate

Consider model M_τ which assumes a lifetime distribution with hazard rate

$$h(t) = \begin{cases} \lambda_1 & \text{for } t \leq \tau \\ \lambda_2 & \text{for } t > \tau \end{cases} \quad (2.1)$$

with $\lambda_1, \lambda_2 > 0$ and $a < \tau < b$. The change-point τ and parameters λ_1 and λ_2 are unknown. Then density function corresponding to the hazard rate in (2.1) is given by

$$f(t | \lambda_1, \lambda_2, \tau) = \begin{cases} \lambda_1 e^{-\lambda_1 t} & , t \leq \tau \\ \lambda_2 e^{-(\lambda_1 - \lambda_2)\tau} e^{-\lambda_2 t} & , t > \tau. \end{cases} \quad (2.2)$$

This is the model considered by most of the references cited in section 1.

Assuming exactly one change-point, Achcar and Bolfarine (1989) consider a discrete prior on τ with the observed values as support, and a vague prior,

$$\pi(\lambda_1, \lambda_2) \propto \frac{1}{\lambda_1 \lambda_2}, \quad \lambda_1, \lambda_2 > 0, \quad (2.3)$$

on the parameters. They also derive the noninformative Jeffreys prior in the case of uncensored and censored observations, assuming τ known, and show that the above prior is a reasonable approximation to the Jeffreys prior.

In Achcar and Loibel (1998) they consider the same model, but obtain the posterior measures by using Gibbs sampling. Ghosh *et al.* (1993, 1996) consider the assumptions one needs to make about the prior in order to obtain

an analytical approximation to the posterior density of τ . The prior in (2.3) would satisfy their conditions if $0 < c < \lambda_1 < \lambda_2 < \infty$ and λ is uniform on $[a, b]$ where $0 < a < \tau < b < \infty$.

We will start by assuming proper conjugate priors on the parameters,

$$\lambda_i \sim \text{Gamma}(\alpha_i, \beta_i), \quad i = 1, 2, \quad (2.4)$$

and a continuous uniform prior for τ on $[a, b]$.

From a sample of $n + m$ observations, let $t_1 < t_2 < \dots < t_n$ be the uncensored and $t_1^* < t_2^* < \dots < t_m^*$ the censored observations. We then take $a = t_1$ and $b = t_{n-1}$. This means we assume that the change-point lies within the range of uncensored observations, an assumption without which the parameters would not be estimable with vague priors. Let k_1 and k_2 be, respectively, the number of uncensored observations before and after the change-point τ . Similarly we have k_1^* and k_2^* for the censored observations.

Further

$$\begin{aligned} s_1 &= y_1 + y_1^* = \sum_{i=1}^{k_1} t_i + \sum_{i=1}^{k_1^*} t_i^*, \\ s_2 &= y_2 + y_2^* = \sum_{i=k_1+1}^n t_i + \sum_{i=k_1^*+1}^m t_i^*, \\ \text{and } r_2 &= k_2 + k_2^*. \end{aligned} \quad (2.5)$$

The posterior density of the change-point is then given by

$$\begin{aligned} \pi(\tau | \mathbf{t}, \mathbf{t}^*) &\propto \Gamma(k_1 + \alpha_1) \Gamma(k_2 + \alpha_2) [s_1 + r_2 \tau + \beta_1]^{-(k_1 + \alpha_1)} \\ &\quad \cdot [s_2 - r_2 \tau + \beta_2]^{-(k_2 + \alpha_2)}, \\ x_k &\leq \tau < x_{k+1}, \end{aligned} \quad (2.6)$$

where $x_1 < x_2 < \dots < x_{n+m}$ are all the ordered observations and k is running from the index of the first to the $(n-2)$ -th uncensored observation. So the posterior of τ has a discontinuity at each observed value of the variable. Equation (2.6) corresponds, in different notation, to the results of Achcar and Bolfarine (1989). However, they assumed a discrete prior for τ . The posterior with vague prior (2.3) follows directly by letting $\alpha_i \rightarrow \beta_i \rightarrow 0$ in (2.6). Also, the posteriors of λ_1 and λ_2 , conditional on τ , follow as

$$\lambda_i | \mathbf{t}, \mathbf{t}^*, \tau \sim \text{Gamma}(k_i, z_i), \quad i = 1, 2, \quad (2.7)$$

where

$$z_1 = s_1 + r_2 \tau \quad \text{and} \quad z_2 = s_2 - r_2 \tau. \quad (2.8)$$

Let $\delta = \frac{\lambda_2}{\lambda_1}$, then

$$\pi(\delta | \mathbf{t}, \mathbf{t}^*, \tau) = \frac{1}{B(k_1, k_2)} \left(\frac{z_2}{z_1} \right)^{k_2 \delta^{k_2-1}} \left(1 + \frac{z_2}{z_1} \delta \right)^{-n}, \quad \delta > 0, \quad (2.9)$$

a Beta Type II distribution, so that $\frac{k_1 z_2}{k_2 z_1} \delta$ has an F-distribution with $2k_1$ and $2k_2$ degrees of freedom. The unconditional distributions of λ_1 , λ_2 and δ follow by averaging over the marginal posterior of τ in (2.6).

In the above analysis the assumption of exactly one change-point in the hazard rate was made. To examine the evidence in favour of this assumption, we will consider the Bayes factor when comparing this model M_τ in (2.2) with the model M_0 which assumes a constant hazard with no change, that is,

$$f(t | \lambda_0) = \lambda_0 e^{-\lambda_0 t}, \quad t > 0. \quad (2.10)$$

The Bayes factor in favour of M_0 is the ratio of the marginal likelihoods of the data under the two models,

$$B_{0,\tau} = \frac{m_0(\mathbf{t}, \mathbf{t}^*)}{m_\tau(\mathbf{t}, \mathbf{t}^*)}, \quad (2.11)$$

where the priors used should in general be proper, as the normalizing constants have an important influence on the ratio. In the absence of proper priors for the parameters, various forms of “partial Bayes factors” have been proposed in the literature. Generally it implies taking part of the information from the data to turn an improper prior into a proper one. See, for example, Spiegelhalter and Smith (1982) and Berger and Pericchi (1996). We will use the fractional Bayes factor proposed by O’Hagan (1995, 1997) where a fraction b ($b < 1$) of the likelihood is combined with an improper prior to produce a proper one. So for a model with likelihood function $L(\theta | \mathbf{x})$ and with improper prior $\pi(\theta)$, the proper prior $\pi^*(\theta)$ follows from

$$\pi^*(\theta) \propto [L(\theta | \mathbf{x})]^b \pi(\theta), \quad (2.12)$$

and the fractional marginal likelihood of the data is then (after normalizing $\pi^*(\theta)$),

$$m^b(\mathbf{x}) = \int [L(\theta | \mathbf{x})]^{1-b} \pi^*(\theta) d\theta. \quad (2.13)$$

So the marginal of the data under model M_0 in (1.11) with vague prior $\pi(\lambda_0) \propto \frac{1}{\lambda_0}$, can be written as

$$m_0^b(\mathbf{t}, \mathbf{t}^*) = \frac{q_0}{q_0^b} \quad (2.14)$$

where

$$\begin{aligned} q_0^b &= \int_0^\infty [L(\lambda_0 | \mathbf{t}, \mathbf{t}^*)]^b \pi(\lambda_0) d\lambda_0 \\ &= \Gamma(nb) b^{-nb} s^{-nb} \end{aligned} \quad (2.15)$$

where

$$s = y + y^* = \sum_{i=1}^n t_i + \sum_{i=1}^m t_i^*. \quad (2.16)$$

Under model M_τ in (2.2) with priors as in (2.3) and with τ uniformly distributed on $[t_1, t_{n-1}]$, we have

$$m_\tau^b(\mathbf{t}, \mathbf{t}^*) = \frac{q_\tau}{q_\tau^b} \quad (2.17)$$

where

$$q_\tau^b = \sum_k \int_{x_k}^{x_{k+1}} \Gamma(bk_1) \Gamma(bk_2) b^{-nb} z_1^{-bk_1} z_2^{-bk_2} d\tau, \quad (2.18)$$

which must be integrated numerically. The quantities z_1 and z_2 are defined in (2.8) and (2.5), and q_0 and q_τ follow by setting $b = 1$ in (2.15) and (2.18).

The fractional Bayes factor in favour of model M_0 is now given by (2.11) and the posterior probability for model M_0 is

$$P[M_0 | \mathbf{t}, \mathbf{t}^*,] = \left[1 + \frac{1-p}{p} B_{0\tau}^{-1} \right]^{-1} \quad (2.19)$$

when p is the prior probability for M_0 .

The suggested value for b is $\frac{n_0}{n}$, where n_0 is the minimal training sample size for the more complex model (O'Hagan, 1995). This leaves the maximal amount of information in the likelihood for model comparison. This means we will take $b = \frac{2}{n}$, where n is the number of uncensored observations, when comparing M_0 with M_τ above.

3. Monotonic Hazard Rates

Firstly we will consider a lifetime distribution with a strictly decreasing hazard rate up to a change-point τ after which it remains constant. Such a model is important in situations where equipment have high infant mortality rate and manufacturers are interested in the point at which the surviving equipment can be considered more reliable.

In general, if $f_1(\cdot)$ and $f_2(\cdot)$ are non-increasing continuous density functions with survival functions $S_1(\cdot)$ and $S_2(\cdot)$ respectively, and

$$f(t|\tau, \theta_1, \theta_2) = \begin{cases} f_1(t|\theta_1) & , \quad t \leq \tau \\ \frac{S_1(\tau|\theta_1)}{S_2(\tau|\theta_2)} f_2(t|\theta_2) & , \quad t > \tau, \end{cases} \quad (3.1)$$

then $h(t_1) \geq h(t_2)$ for $t_1 < t_2$ if $\theta_2 = g(\theta_1)$ such that

$$f_1(\tau|\theta_1)S_2(\tau|\theta_2) = f_2(\tau|\theta_2)S_1(\tau|\theta_1).$$

This last condition defines the assumption that there is no jump in the density or hazard functions at the point of discontinuity.

We will assume a Weibull density with decreasing hazard rate for $f_1(\cdot)$ and an exponential density for $f_2(\cdot)$.

Then

$$f(t|\tau, \lambda, \beta) = \begin{cases} \lambda\beta t^{\beta-1}e^{-\lambda t^\beta} & , \quad t \leq \tau \\ \lambda\beta\tau^{\beta-1}e^{-\lambda\tau^{\beta-1}[(1-\beta)\tau+\beta t]} & , \quad t > \tau \end{cases} \quad (3.2)$$

with hazard rate

$$h(t) = \begin{cases} \lambda\beta t^{\beta-1} & , \quad t \leq \tau \\ \lambda\beta\tau^{\beta-1} & , \quad t > \tau, \end{cases} \quad (3.3)$$

where $\lambda > 0$ and $0 < \beta < 1$.

The likelihood function is given by

$$\begin{aligned} L(\lambda, \beta, \tau | \mathbf{t}, \mathbf{t}^*) &= \lambda^n \beta^n \tau^{-k_2(1-\beta)} \left(\prod_{i=1}^{k_1} t_i \right)^{\beta-1} \\ &\quad \times e^{-\lambda[s_{1\beta} + \tau^{\beta-1}(1-\beta)\tau + \beta\tau^{\beta-1}s_2]} \end{aligned} \quad (3.4)$$

with notation as in (2.5) and

$$s_{1\beta} = \sum_{i=1}^{k_1} t_i^\beta + \sum_{i=1}^{k_1^*} t_i^{*\beta}. \quad (3.5)$$

With uniform priors on β and τ , and $\pi(\lambda) \propto \frac{1}{\lambda}$, the joint posterior of β and τ follows as

$$\pi(\beta, \tau | \mathbf{t}, \mathbf{t}^*) \propto \left(\tau^{k_2} \prod_{i=1}^{k_1} t_i \right)^{\beta-1} \left[\frac{s_1 \beta}{\beta} + \left(\frac{1-\beta}{\beta} \right) r_2 \tau^\beta + \tau^{\beta-1} s_2 \right]^{-n}, \quad (3.6)$$

$$0 < \beta < 1, \quad t_1 \leq \tau \leq t_{n-1}.$$

The marginal posteriors can be obtained by numerical integration or Gibbs sampling. Let the model (3.2) be denoted by M_τ , while M_0 denotes a single Weibull distribution with decreasing hazard rate. The fractional Bayes factor for comparison of these two models follows as in (2.11) and (2.14), where

$$q_0^b = \int_0^1 \beta^{bn} \left(\prod_{i=1}^n t_i \right)^{b(\beta-1)} s_\beta^{-bn} d\beta \quad (3.7)$$

with

$$s_\beta = \sum_{i=1}^n t_i + \sum_{i=1}^m t_i^*, \quad (3.8)$$

and q_τ^b is the integrated value of (3.6) to the power b . An MCMC method of obtaining the fractional Bayes factor will be discussed in the next section.

In the situation where an initially constant hazard rate later changes to an increasing rate due to wear-out or ageing, we have a combination of an exponential distribution and a Weibull distribution with increasing hazard rate.

Then

$$f(t | \lambda, \beta, \tau) = \begin{cases} \lambda e^{-\lambda t} & , \quad t \leq \tau \\ \lambda e^{-\lambda \tau \left(\frac{\beta-1}{\beta} \right)} \left(\frac{t}{\tau} \right)^{\beta-1} e^{-\frac{\lambda \tau}{\beta} \left(\frac{t}{\tau} \right)^\beta} & , \quad t > \tau \end{cases} \quad (3.9)$$

with hazard rate

$$h(t) = \begin{cases} \lambda & , \quad t \leq \tau \\ \lambda \left(\frac{t}{\tau} \right)^{\beta-1} & , \quad t > \tau \end{cases} \quad (3.10)$$

where $\beta > 1$.

In this case we need a bounded parameter space for β with a vague prior as the likelihood function is unbounded. So let

$$\pi(\beta) \propto \frac{1}{\beta}, \quad 1 \leq \beta < K, \quad (3.11)$$

with the same priors as before on λ and τ .

The joint posterior of τ and β is now

$$\pi(\tau, \beta | t, t^*) \propto \frac{1}{\beta} \tau^{-k_2(\beta-1)} \left(\prod_{i=k_1+1}^n t_i \right)^{\beta-1} \left[s_1 + \tau r_2 \left(\frac{\beta-1}{\beta} \right) + \frac{\tau^{1-\beta}}{\beta} y_2 \beta \right]^{-n},$$

$$t_1 \leq \tau < t_{n-1}, \quad 1 \leq \beta < K. \quad (3.12)$$

Comparison with model M_0 in (2.10) with q_0^b given in (2.15) can now be made.

4. Non-monotonic Hazard Rate

A lifetime distribution having support $[0, \infty)$ is said to be a Bathtub Hazard Rate (BHR) distribution if there exists a $t_0 > 0$ such that $h(t)$ is non-increasing on $[0, t_0)$ and non-decreasing on $[t_0, \infty)$.

The point t_0 need not be unique, and we will consider a model with decreasing failure rate on $[0, \tau_1]$, a constant failure rate on $(\tau_1, \tau_2]$, and an increasing failure rate on (τ_2, ∞) . Assuming no jumps at the points of discontinuity, the density function can be written as

$$f(t | \lambda, \beta_1, \beta_2, \tau_1, \tau_2) = \begin{cases} \lambda \beta_1 t^{\beta_1-1} e^{-\lambda t^{\beta_1}} & , \quad 0 \leq t \leq \tau_1 \\ \lambda \beta_1 \tau_1^{\beta_1-1} e^{-\lambda \tau_1^{\beta_1} (1-\beta_1)} e^{-\lambda \beta_1 \tau_1^{\beta_1-1} t} & , \quad \tau_1 < t \leq \tau_2 \\ A t^{\beta_2-1} e^{-B t^{\beta_2}} & , \quad t_2 < t < \infty \end{cases} \quad (4.1)$$

with $\lambda > 0$, $0 < \beta_1 \leq 1$, $1 \leq \beta_2 \leq K$, and

$$\begin{aligned}
 A &= \beta_2 B e^{-\lambda \beta_1 \frac{\beta_2 - 1}{\beta_2} \tau_1^{\beta_1 - 1} \tau_2 - \lambda \tau_1^{\beta_1} (1 - \beta_1)}, \\
 B &= \lambda \frac{\beta_1}{\beta_2} \frac{\tau_1^{\beta_1 - 1}}{\tau_2^{\beta_2 - 1}}.
 \end{aligned} \tag{4.2}$$

The corresponding hazard rate is

$$h(t) = \begin{cases} \lambda \beta_1 t^{\beta_1 - 1} & , \quad 0 \leq t \leq \tau_1 \\ \lambda \beta_1 \tau_2^{\beta_1 - 1} & , \quad \tau_1 < t \leq \tau_2 \\ \beta_2 B t^{\beta_2 - 1} & , \quad \tau_2 < t < \infty. \end{cases} \tag{4.3}$$

The priors are

$$\begin{aligned}
 \pi(\lambda) &\propto \frac{1}{\lambda}, \quad \pi(\beta_1) = 1, \quad 0 < \beta_1 \leq 1, \\
 \pi(\beta_2) &\propto \frac{1}{\beta_2}, \quad 1 \leq \beta_2 \leq K, \\
 \pi(\tau_2 | \tau_1) &= U(\tau_1, b), \quad \pi(\tau_1) = U(a, b).
 \end{aligned} \tag{4.4}$$

In practice, under the assumption that both change-points fall within the range of observed data, we will take $a = t_1$ and b as either t_{n-1} or t_{n-2} , ensuring at least one observation in each section of the density function.

To obtain the marginal posterior distributions, Gibbs sampling can be employed, simulating from the conditional distributions successively and averaging. After integrating λ out, the full conditional distributions of the other four parameters are given by

$$\pi(\beta_1 | \mathbf{t}, \mathbf{t}^*, \beta_2, \tau_1, \tau_2) \propto \beta_1^n \tau_2^{\beta_1(k_2 + k_3)} T_1^{\beta_1 - 1} C^{-n}, \quad 0 < \beta_1 \leq 1, \tag{4.5}$$

$$\pi(\beta_2 | \mathbf{t}, \mathbf{t}^*, \beta_1, \tau_1, \tau_2) \propto \frac{1}{\beta_2} \tau_2^{-k_3(\beta_2 - 1)} T_3^{\beta_2 - 1} C^{-n}, \quad 1 \leq \beta_2 \leq K, \tag{4.6}$$

$$\pi(\tau_1 | \mathbf{t}, \mathbf{t}^*, \beta_1, \beta_2, \tau_2) \propto \tau_1^{-(k_2 + k_3)(\beta_1 - 1)} T_1^{\beta_1 - 1} C^{-n}, \quad t_1 \leq \tau_1 < t_{[\tau_2]} \tag{4.7}$$

$$\pi(\tau_2 | \mathbf{t}, \mathbf{t}^*, \beta_1, \beta_2, \tau_1) \propto \tau_2^{-k_3(\beta_2 - 1)} T_3^{\beta_2 - 1} C^{-n}, \quad t_{[\tau_2] +} < \tau_2 \leq t_{n-1}, \tag{4.8}$$

where k_1 , k_2 and k_3 are the number of observations smaller or equal to τ_1 , between τ_1 and τ_2 , and larger than τ_2 respectively. Similarly we have k_1^* , k_2^* and k_3^* for censored observations.

Further

$$T_1 = \prod_{i=1}^{k_1} t_i, \quad T_3 = \prod_{i=k_1+k_2+1}^n t_i, \quad (4.9)$$

and

$$C = (r_2 + r_3) \tau_1^{\beta_1} (1 - \beta_1) + \beta_1 \tau_1^{\beta_1 - 1} \left(r_3 \frac{\beta_2 - 1}{\beta_2} \tau_2 + s_2 + \frac{1}{\beta_2} \tau_2^{1 - \beta_2} s_{3\beta_2} \right) + s_{1\beta_1}, \quad (4.10)$$

where $r_i = k_i + k_i^*$, $i = 1, 2, 3$, and

$$s_{1\beta_1} = \sum_{i=1}^{r_1} x_i^{\beta_1}, \quad s_2 = \sum_{i=r_1+1}^{r_1+r_2} x_i, \quad s_{3\beta_2} = \sum_{i=r_1+r_2+1}^{n+m} x_i^{\beta_2}, \quad (4.11)$$

where $x_1 < x_2 < \dots < x_{n+m}$ are again all the ordered observations.

The conditional distribution of λ is $\text{Gamma}(n, C)$. For model comparisons, the fractional marginal likelihood as in (2.14) can be obtained by an MCMC sampling scheme. As mentioned by Gilks (1995), notice from (2.13) that

$$m^b(\mathbf{x}) = E \left[L^{1-b}(\theta|\mathbf{x}) \right], \quad (4.12)$$

with expectation with respect to the proper density $\pi^*(\theta)$ given in (2.12). So we perform MCMC iterations on the density (2.12) and estimate $m^b(\mathbf{x})$ as the average of $L^{1-b}(\theta|\mathbf{x})$ over the generated samples. The same method can be employed for the models in section 3.

5. Applications

Example 1

We will consider a data set (Appendix A) given in Matthews and Farewell (1982) as well as in Achcar and Bolfarine (1989) and Achcar and Loibel (1998).

The data specifies the times (in days) from remission induction to relapse for 84 patients with acute nonlymphoblastic leukemia. There are 51 uncensored and 33 censored observations.

The fractional Bayes factor in favour of model M_0 (no change) when compared with model M_τ is $B_{0\tau} = 0.0628$ when $b = \frac{2}{n}$. This translates to a posterior probability for the change-point model of $P = 0.9409$ with equal prior weights. This indicates reasonably strong evidence in favour of the change-point model. In figure 1 the fractional Bayes factor is shown as a function of b . As b approaches one, so will $B_{0\tau}$, as there is then little or no information left for model comparison. However, for b reasonably small ($b < 0.5$) the Bayes factor is fairly robust ($B_{0\tau} < 0.1$) with a minimum of 0.0367 at $b = \frac{8}{n}$.

According to Matthews and Farewell (1982) the classical significance level when testing the null hypothesis of no change against the change-point alternative is approximately 0.001.

We also compared the change-point model M_τ with a Weibull model with decreasing hazard rate. The fractional Bayes factor in favour of M_τ is 53.64, indicating a much better fit to the data than a Weibull model.

Assuming a change-point, figure 2 shows the posterior distribution of τ . The discontinuities occur at each of the uncensored observations.

The mode is at 697 days. This is also the maximum likelihood estimator of τ (Matthews and Farewell, 1982). A HPD region will be a set of disjointed intervals and not of much practical use. We can state that the probability that the change-point lies between the 49th and 50th uncensored observations (697 to 955 days) is 0.614.

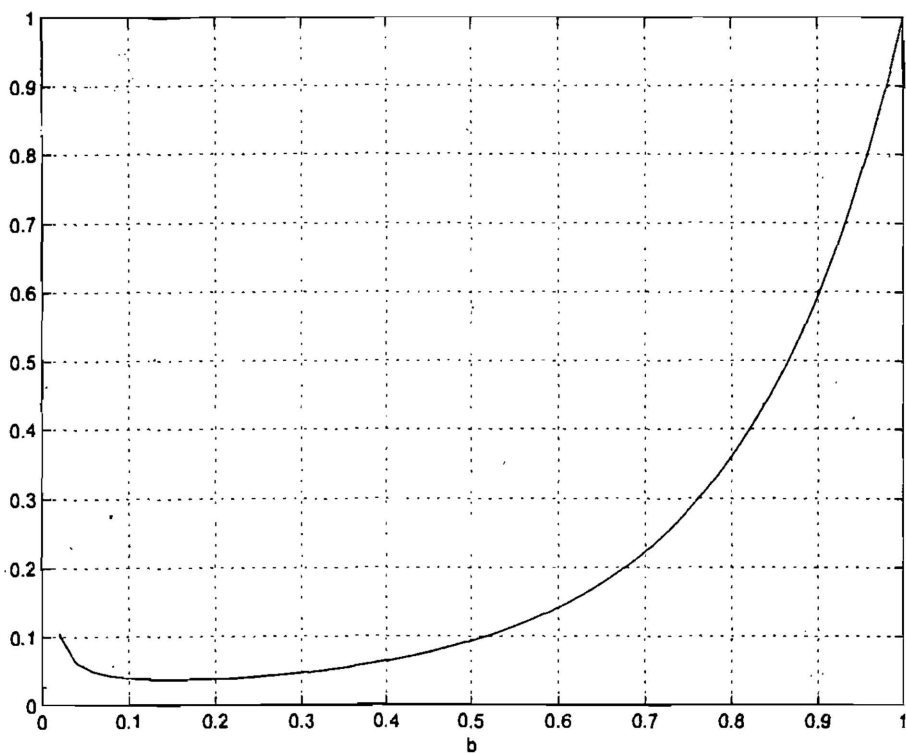


Figure 1: The Fractional Bayes Factor as a function of b

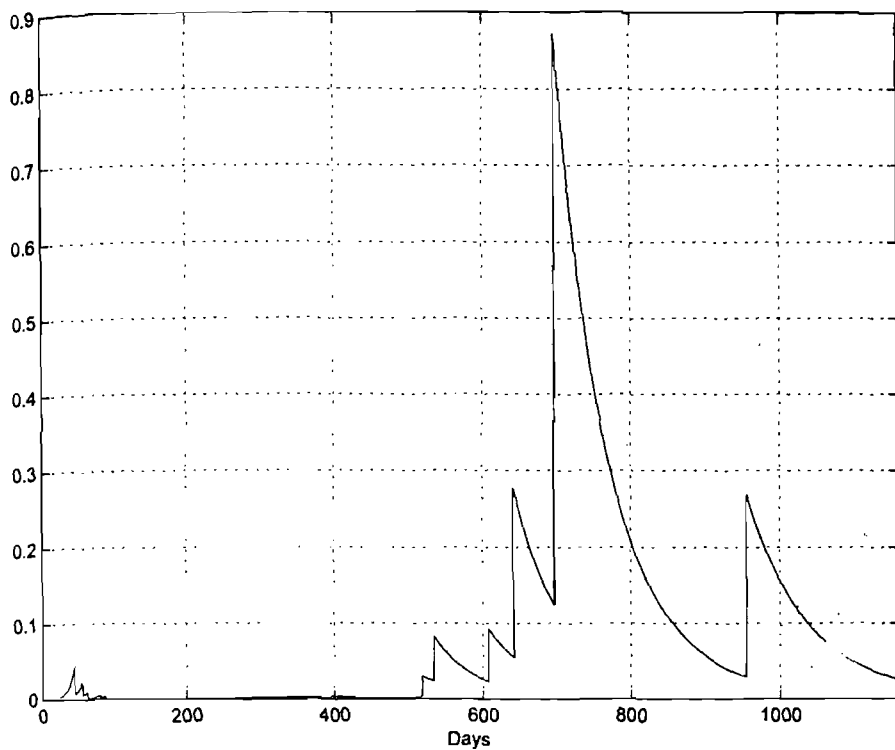


Figure 2 : The posterior distribution of the change-point τ

The unconditional posterior means of λ_1 and λ_2 are given in Table 1 together with the asymptotic approximations according to Ghosh *et al.* (1996) (GJM), the maximum likelihood estimators from Matthews and Farewell (1982) (MF), and the results of Achcar and Loibel (1998) (AL), using Gibbs sampling.

Table 1: Estimates of λ_1 and λ_2

	$\lambda_1 \times 10^3$	$\lambda_2 \times 10^3$
Post. Means	2.005	0.3364
Approximation (GJM)	2.081	0.2877
MLE (MF)	2.040	0.4300
Gibbs (AL)	2.000	0.3204

In figure 3 the conditional, given $\tau = 697$, and unconditional posteriors of $\delta = \frac{\lambda_2}{\lambda_1}$ is shown according to (2.9).

The unconditional posterior of δ has a mean of 0.1512 and 95% HPD interval (0.007;0.393). The maximum likelihood estimator is $\hat{\delta} = 0.2108$.

Assuming both τ and λ_1 known, and equal to the MLEs, Achcar and Bolfarine (1989) report the posterior mode of δ as 0.1405, and with only τ known, as 0.1374. They also give in this case a 95% HPD interval of (0.1253;0.3540) by using a normal approximation. We consider this interval as too narrow. Their unconditional posterior mode is 0.25. In Achcar and Loibel (1998) a posterior mean of 0.1602 and 95% HPD interval of (0.0209 ; 0.4550) is reported, which seems more reasonable and closer to our results.

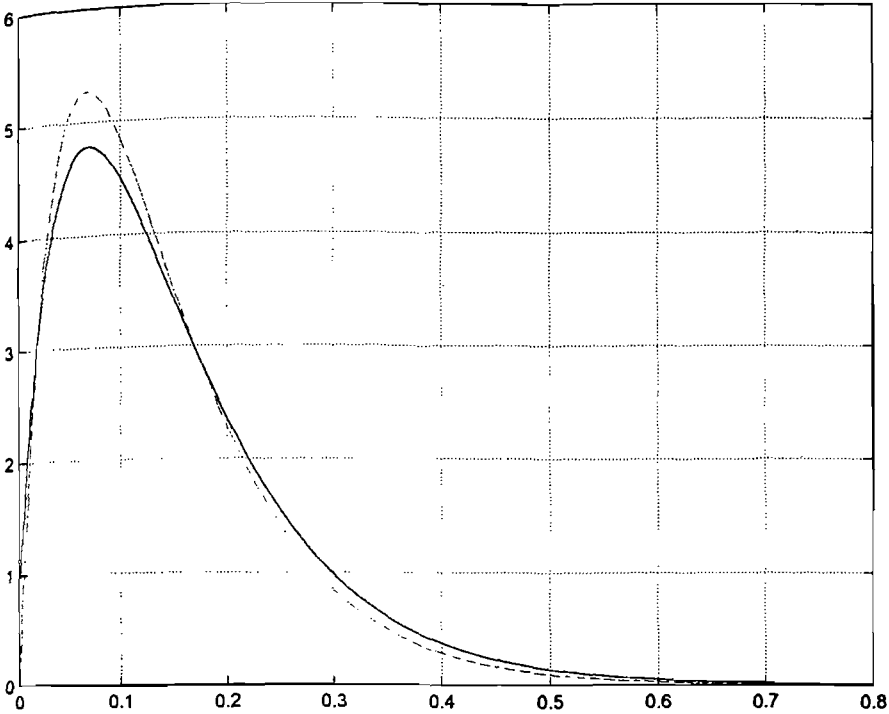


Figure 3: Conditional (given $\tau = 697$) posterior (---) and unconditional posterior (—) of $\delta = \lambda_2/\lambda_1$

A refinement to the analysis in section 2 is possible when a change-point is assumed to exist and it is known that $\lambda_2 < \lambda_1$. The vague prior (2.3) is then replaced by

$$\pi(\lambda_1, \lambda_2) \propto \frac{1}{\lambda_1 \lambda_2}, \quad 0 < \lambda_2 < \lambda_1 < \infty.$$

The posterior of τ is then not analytically tractable, but is given by

$$\pi(\tau | t, t^*) \propto z_2^{-k_2} \int_0^\infty \Gamma(\lambda_1 z_2; k_2) \lambda_1^{k_1-1} e^{-\lambda_1 z_1} d\lambda_1 \quad (5.2)$$

where $\Gamma(\cdot; k_2)$ is the incomplete Gamma function with parameter k_2 , and can readily be evaluated numerically.

Under the assumption that $\lambda_2 < \lambda_1$, the probability of the change-point between 697 and 955 days increases to 0.699. The posterior mean of δ is 0.1411 with 95% HPD interval (0.003; 0.342).

Example 2

In Appendix B the failure times (hours) of 107 units of a piece of electronic equipment are given. The data is from Juron and Gryna (1980) and is also analysed by Schneider *et al.* (1990). It is suggested by Schneider *et al.* that there is an early failure period of approximately 20 hours and a wear-out period of about 100 hours, the period between being fairly constant.

We compared the non-decreasing hazard rate model (3.10) with the bathtub model (4.3) by means of the MCMC calculation (4.12) of the fractional Bayes factor. After 10000 simulations from each model and with $b = 3/n$, the estimated Bayes factor was calculated as $B = 3.79$ in favour of model (4.3), moderate evidence in favour of the bathtub hazard rate model. Compared to a constant hazard rate model and the single abrupt change model (2.1), the Bayes factors in favour of the BHR model are, respectively, 1.2×10^9 and 1.3×10^3 .

Assuming the BHR model, 15000 Gibbs samples were kept after burn-in, and the marginal posterior densities of the five parameters obtained. Some posterior measures are given in Table 2 with the two estimated change-points at 46.9 and 95.5 hours. The position of the second change-point, τ_2 , can be estimated much more accurately than that of τ_1 , as can be seen from the 95% HPD Interval. Also some contours of the joint density of τ_1 and τ_2 are shown in figure 4.

Table 2: Posterior measures of five parameters

	Mean	Mode	Variance	95% Interval	
λ	0.2071	0.1990	0.0015	0.137	– 0.289
β_1	0.7763	0.7780	0.0081	0.604	– 0.958
β_2	11.8259	10.6300	23.8981	4.450	– 20.400
τ_1	46.922	26.100	483.37	10.80	– 89.40
τ_2	95.546	97.100	82.99	78.70	– 112.80

Figure 5 shows the expectation of the unconditional distribution of the hazard rate (4.3) with the 95% credibility interval, and in figure 6 the empirical survival function $S(t) = 1 - F(t)$ is plotted together with the posterior expectation and 95% credibility interval.

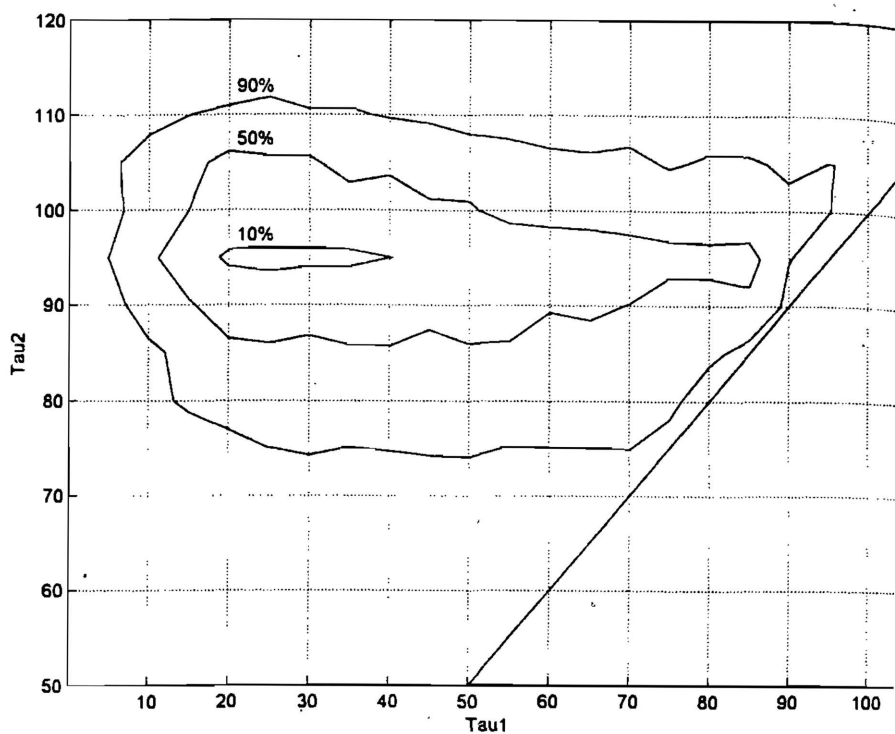


Figure 4: Contours of the joint posterior density of the change-points τ_1 and τ_2

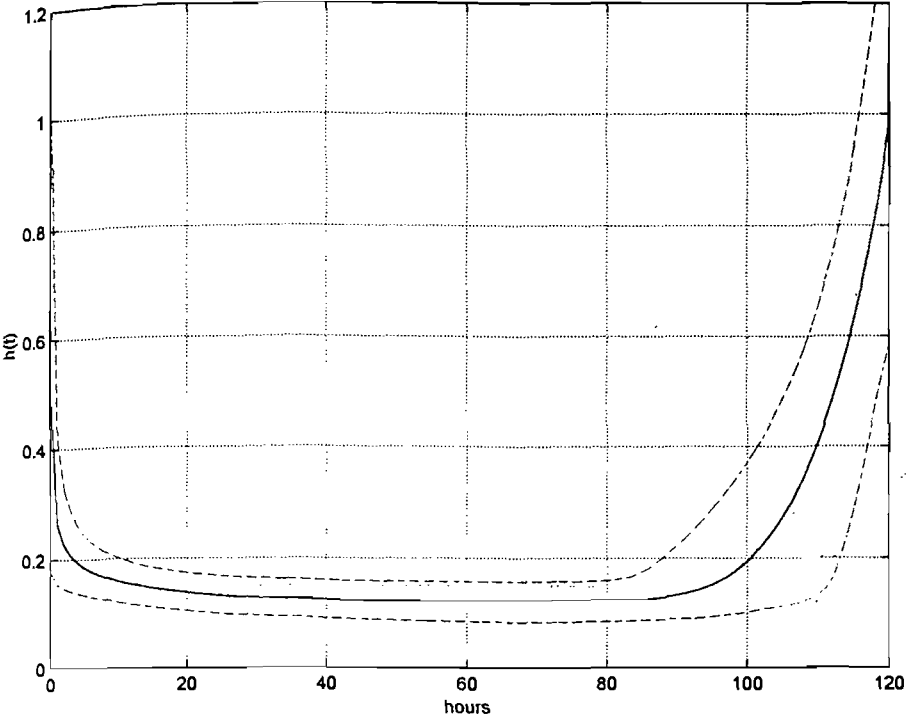


Figure 5: Estimated Hazard Rate with 95% Credibility Interval

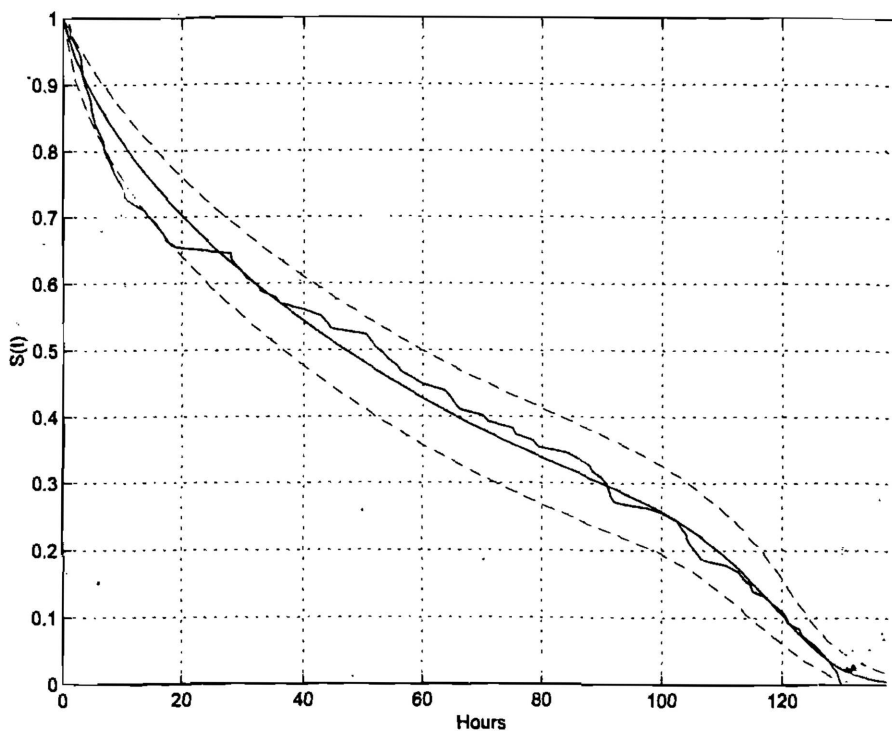


Figure 6: Empirical and Estimated Survival Function with 95% Credibility Interval

6. Conclusions

In this study we first considered the detection of a single abrupt change in an otherwise constant hazard rate from a Bayesian point of view. A number of Bayesian papers dealing with this subject have appeared in the literature, with the emphasis on the estimation of the position of the change-point and other model parameters. The question of the existence of a change-point has largely been ignored. In this analysis the fractional Bayes factor is used as a measure of evidence in favour of the change-point model in the absence of any prior information.

Monotonic and non-monotonic hazard rates are also considered and modelled as combinations of exponential and Weibull lifetime distributions with one or two change-points. It is often important to determine whether a component has a higher hazard rate due to infancy, old age or both, and model comparison can again be done by use of the fractional Bayes factor. We believe that the fractional Bayes factor can be an important tool in determining the best hazard rate model and a Monte Carlo method of calculating it is also illustrated.

References

- ACHCAR, J.A. & BOLFARINE, H. (1989). Constant Hazard against a Change-point alternative: A Bayesian Approach with Censored Data. *Commun. Statist.-Theor. Meth.*, **18**, 3801–3819.
- ACHCAR, J.A. & LOIBEL, S. (1998). Constant Hazard function Models with a Change Point: A Bayesian Analysis using Markov Chain Monte Carlo Methods. *Biometrical Journal*, **40**, 543–555.
- BASU, A.P., GHOSH, J.K. & JOSHI, S.N. (1988). On Estimating Change-point in a Hazard Rate, *Statistical Decision theory and Related Topics*, (S.S. Gupta and J.O. Berger eds.) Springer-Verlag, New York, 239–252.

- BERGER, J.O. & PERICCHI, L. (1996). The intrinsic Bayes factor for model selection and prediction. *J. Amer. Statist. Assoc.*, **91**, 109–122.
- GHOSAL, S., GHOSH, J.K. & SAMANTA, T. (1999). Approximation of the Posterior Distribution in a Change-point Problem. *Ann. Inst. Statist. Math.*, **51**, 479–497.
- GHOSH, J.K. & JOSHI, S.N. (1992). On the Asymptotic Distribution of an Estimate of the Change-point in a Failure Rate. *Commun. Statist.-Theor. Meth.*, **21**, 3571–3588.
- GHOSH, J.K., JOSHI, S.N. & MUKHOPADHYAY, C. (1993). A Bayesian Approach to the Estimation of the Change-point in a Hazard Rate. *Advances in Reliability*, (A.P. Basu ed.) Elsevier Science Publishers, North-Holland, 141–170.
- GHOSH, J.K., JOSHI, S.N. & MUKHOPADHYAY, C. (1996). Asymptotics of a Bayesian Approach to Estimating Change-point in a Hazard Rate. *Commun. Statist.-Theor. Meth.*, **25**, 3147–3166.
- GILKS, W.R. (1995). Discussion of the paper by O'Hagan. *J. Roy. Statist. Soc. B*, **97**, 118–120.
- JURAN, J. & GRZYNA, F. (1980). *Quality planning and analysis*. McGraw-Hill, New York.
- KULASEKERA, K.B. & SAXENA, K.M.L. (1991). Estimation of Change Point in Failure Rate Models. *J. Statist. Plann. Inference*, **29**, 111–124.
- MATTHEWS, D.E. & FAREWELL, V.T. (1982). On testing for a Constant Hazard against a Change-point Alternative. *Biometrics*, **38**, 463–468.
- MATTHEWS, D.E., FAREWELL, V.T. & PYKE, R. (1985). Asymptotic Score Statistic Processes and Tests for Constant Hazard against a Change-point Alternative. *Ann. Statist.*, **13**, 583–591.
- MITRA, M. & BASU, S.K. (1995). Change-point Estimation in Non-monotonic Ageing Models. *Ann. Inst. Statist. Math.*, **47**, 483–491.
- NGUYEN, H.T., ROGERS, G.S. & WALKER, E.A. (1984). Estimation in Change-point Hazard Rate Models. *Biometrika*, **71**, 299–304.

- O'HAGAN, A. (1995). Fractional Bayes factors for model comparison. *J. Roy. Statist. Soc. B*, **57**, 99–138.
- O'HAGAN, A. (1997). Properties of intrinsic and fractional Bayes factors. *Test*, **6**, 101–118.
- PHAM, T.D. & NGUYEN, H.T. (1990). Strong consistency of the Maximum Likelihood Estimators in the Change-point Hazard Rate Model, *Statistics*, **21**, 2, 203–216.
- PHAM, T.D. & NGUYEN, H.T. (1993). Bootstrapping the Change-point of a Hazard Rate. *Ann. Inst. Statist. Math.*, **45**, 331–340.
- SCHNEIDER, H., LIN, B-S. & O'CONNOR, C. (1990). Comparisons of nonparametric estimators for the renewal function. *Applied Statistics*, **39**, 55–61.
- SPIEGELHALTER, D.J. & SMITH, A.F.M. (1982). Bayes factors for linear and log-linear models with vague prior information. *J. Roy. Statist. Soc. B*, **44**, 372–387.
- YAO, Y-C. (1986). Maximum Likelihood Estimation in Hazard Rate Models with Change-point. *Commun. Statist.-Theor Meth.*, **15**, 2455–2466.
- ZACKS (1972). On detecting the epoch at which the failure rate of a reliability system starts to increase. Tech. Memo., Serial TM-80320, Program in Logistics, George Washington University.

Appendix A:

*Ordered remission durations for 84 patients with acute
nonlymphoblastic leukemia (Matthews and Farewell, 1982)*

Uncensored observations (51)

24	46	57	57	64	65	82	89
90	90	111	117	128	143	148	152
166	171	186	191	197	209	223	230
247	249	254	258	264	269	270	273
284	294	304	304	332	341	393	395
487	510	516	518	518	534	608	642
697	955	1160					

Censored observations (33)

68	119	182	182	182	182	182	182
182	182	182	182	182	182	182	182
182	182	182	182	182	182	182	182
182	182	583	1310	1538	1634	1908	1996
2057							

Appendix B:

Failure times for a piece of electronic equipment, Juran and Gryna (1980)

1.0	6.4	19.2	54.2	88.4	114.98
1.2	6.8	28.1	55.6	89.9	115.1
1.3	6.9	28.2	56.4	90.8	117.4
2.0	7.2	29.0	58.3	91.1	118.3
2.4	7.9	29.9	60.2	91.5	119.7
2.9	8.3	30.6	63.7	92.1	120.6
3.0	8.7	32.4	64.6	97.9	121.0
3.1	9.2	33.9	65.3	100.8	122.9
3.3	9.8	35.3	66.2	102.6	123.3
3.5	10.2	36.1	70.1	103.2	124.5
3.8	10.4	40.1	71.0	104.0	125.8
4.3	11.9	42.8	75.1	104.3	126.6
4.6	13.8	43.7	75.6	105.0	127.7
4.7	14.4	44.5	78.4	105.8	128.4
4.8	15.6	50.4	79.2	106.5	129.2
5.2	16.2	51.2	84.1	110.7	129.5
5.4	17.0	52.0	86.0	112.6	129.9
5.9	17.5	53.3	87.9	113.5	